

4.4 Overall end-to-end System flow

PrismSpace Architecture

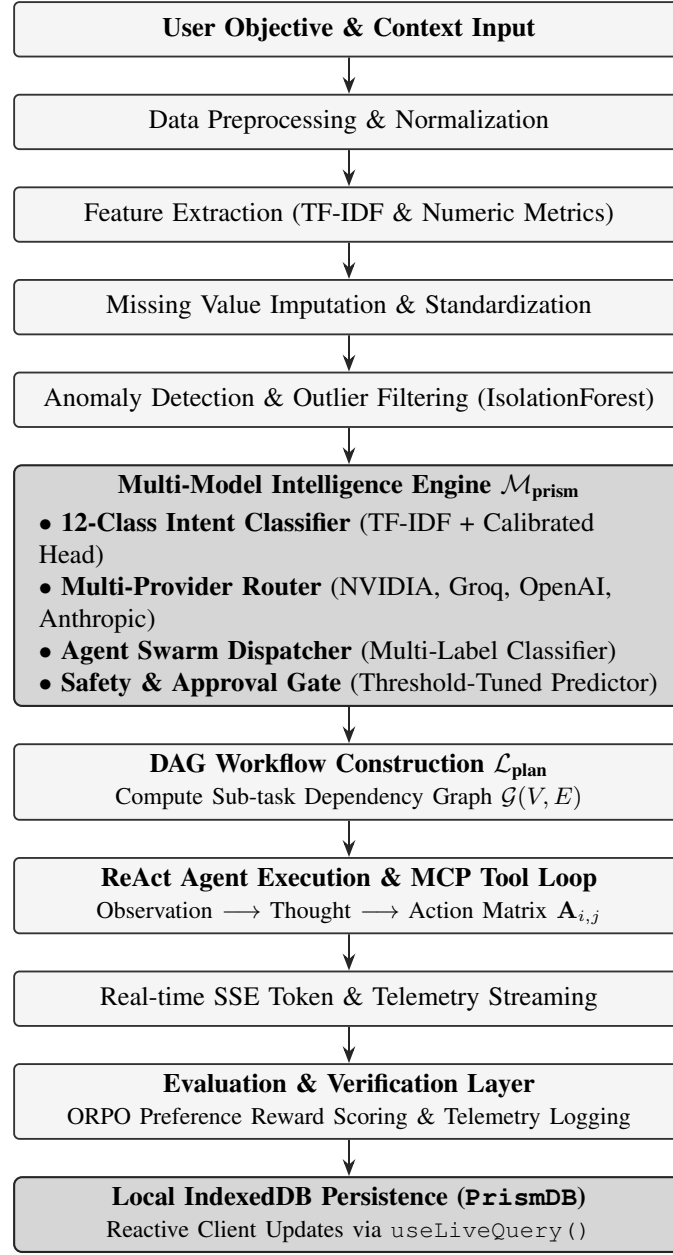


Figure 4.3: System flow

System Flow Glossary & Terminology Reference

TF-IDF: Term Frequency–Inverse Document Frequency; statistical metric evaluating token relevance in user queries.

IsolationForest: Unsupervised decision tree ensemble identifying anomalous or out-of-distribution user inputs.

$\mathcal{M}_{\text{prism}}$: Multi-Model Intelligence Engine; dynamically routes sub-tasks across Groq, NVIDIA, OpenAI, etc.

Safety & Approval Gate: Calibrated classifier halting execution for human approval on high-risk operations.

DAG $\mathcal{G}(V, E)$: Directed Acyclic Graph; orders execution plan where vertices V are tasks and edges E are dependencies.

ReAct: Reasoning and Acting; loop alternating explicit chain-of-thought rationale with concrete tool actions.

MCP: Model Context Protocol; open protocol enabling LLMs to safely invoke local/remote tools and data sources.

SSE: Server-Sent Events; HTTP streaming standard pushing real-time LLM token streams and execution telemetry.

ORPO: Odds Ratio Preference Optimization; reference-free preference alignment optimizing response quality.

PrismDB / `useLiveQuery`: Client-side IndexedDB database offering reactive, zero-latency UI re-rendering on updates.